

Newtonian Fluids

the backend needs some clean-up. didn't finish to rewrite 2023 notes!

A fluid is incompressible Newtonian if:

1) Reference mass density uniform: $\rho_0(x) = \rho_0$

2) Fluid is incompressible $\nabla_x \cdot \underline{v} = 0$

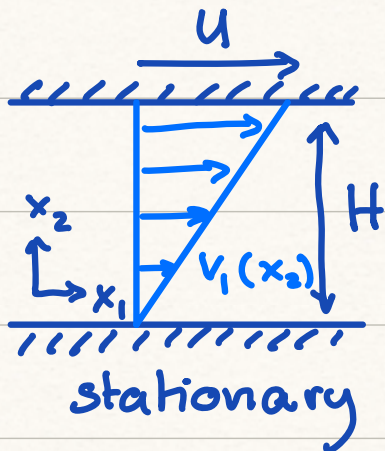
$$\text{Prop 1 + 2} \Rightarrow \rho(x, t) = \rho_0 > 0$$

3) Cauchy stress field is Newtonian

$$\underline{\underline{\sigma}} = \underline{\underline{\sigma}}^h + \underline{\underline{\sigma}}^a$$

$$\text{hydrost. stress: } \underline{\underline{\sigma}}^h = -p \underline{\underline{I}}$$

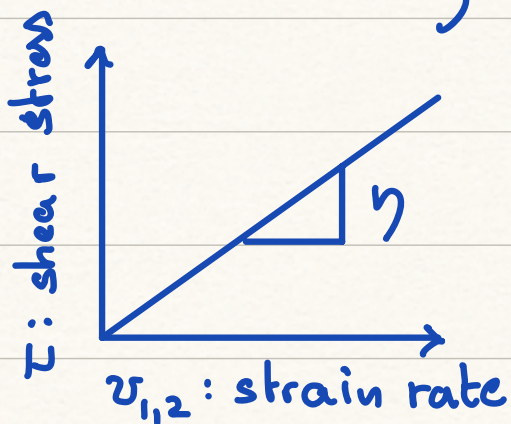
Active stress: $\Rightarrow$ Newton's law of viscosity



Shear force on top plate

$$F = \mu A \frac{U}{H} \quad A = \text{area of plate}$$

μ = dynamic/absolute viscosity
units $\left[\frac{N}{LT}\right]$ eg. Pa s



$$\text{shear stress: } \tau = \frac{F}{A}$$

$$\tau = \mu \frac{\partial v_1}{\partial x_2} \quad \text{Newton's law of viscosity}$$

Generalization to 3D

$$\sigma_{ij}^a = \mu \frac{\partial v_i}{\partial x_j} \Rightarrow \underline{\underline{\sigma}}^a = \mu \nabla \underline{\underline{v}}$$

$$\text{but } \nabla \underline{\underline{v}} \text{ not objective} \Rightarrow \underline{\underline{\sigma}}^a = 2\mu \text{sym}(\nabla \underline{\underline{v}}) \\ = \mu (\nabla \underline{\underline{v}} + \nabla \underline{\underline{v}}^T)$$

$\Rightarrow$ tensor version of classical linear laws

Fourier's law: $\underline{\underline{q}} = -\kappa \nabla T$ (heat conduction)

Fick's law: $\underline{\underline{j}} = -D \nabla C$ (species diffusion)

Newton's law: $\underline{\underline{\sigma}} = \mu \text{sym}(\nabla \underline{\underline{v}})$ (momentum diffusion)

Cauchy stress in Newtonian fluid:

$$\underline{\underline{\sigma}} = -p \underline{\underline{I}} + \mu (\nabla \underline{\underline{v}} + \nabla \underline{\underline{v}}^T)$$

In limit $\mu \rightarrow 0$ Newtonian fluid reduces to ideal fluid.

Compare to Representation Theorem with $\underline{\underline{E}} = \nabla \underline{\underline{v}}$:

$$\underline{\underline{G}}(\underline{\underline{E}}) = \lambda \text{tr}(\underline{\underline{E}}) \underline{\underline{I}} + 2\mu \text{sym}(\underline{\underline{E}})$$

$$\underline{\underline{G}}(\underline{\underline{E}}) = \lambda (\cancel{\nabla \underline{\underline{v}}}) \underline{\underline{I}} + \mu (\nabla \underline{\underline{v}} + \nabla \underline{\underline{v}}^T) = \underline{\underline{\sigma}}^a$$

Navier - Stokes Equations

Setting $\rho = \rho_0$ and $\underline{\underline{\sigma}} = -p \underline{\underline{I}} + \mu (\nabla \underline{v} + \nabla \underline{v}^T)$
we obtain lin. mom. balance

$$\begin{aligned}\rho \dot{\underline{v}} &= \nabla \cdot [-p \underline{\underline{I}} + \mu (\nabla \underline{v} + \nabla \underline{v}^T)] + \rho \underline{g} \\ &= -\nabla p + \nabla \cdot [\mu (\nabla \underline{v} + \nabla \underline{v}^T)]\end{aligned}$$

$$\nabla \cdot (p \underline{\underline{I}}) = (p \delta_{ij})_{,j} \underline{e}_i = (p_{,j} \delta_{ij} \underline{e}_j + p \cancel{\delta_{ij,j}} \underline{e}_i) = p_{,j} \underline{e}_j = \nabla p$$

assuming $\mu = \text{constant}$ we have

$$\mu \nabla \cdot (\nabla \underline{v} + \nabla \underline{v}^T) = \mu \nabla^2 \underline{v} + \mu \nabla \cdot \nabla \underline{v}^T$$

Last term

$$\begin{aligned}\nabla \cdot \nabla \underline{v}^T &= v_{j,i,j} \underline{e}_i = v_{j,i,j} \underline{e}_i = \nabla (\nabla \cdot \underline{v}) \\ \Rightarrow \nabla \cdot \underline{\underline{\sigma}} &= -\nabla p + \mu \nabla^2 \underline{v}\end{aligned}$$

expanding mat. deriv.: $\dot{\underline{v}} = \frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \underline{v}$

$$\rho \left[\frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \underline{v} \right] = \mu \nabla^2 \underline{v} - \nabla p + \rho \underline{g}$$

$$\nabla \cdot \underline{v} = 0$$

Convective form of Navier-Stokes Equation

Scaling Navier Stokes Equations

$$\rho \frac{\partial \underline{u}}{\partial t} + (\nabla \underline{u}) \underline{u} = \eta \nabla^2 \underline{u} - \nabla p + \rho g$$

reduced pressure: $\pi = p + \rho g z$

$$-\nabla p + \rho g = -\nabla p - \rho g \hat{z} = -\nabla (p + \rho g z) = -\nabla \pi$$

we have

$$\rho \left(\frac{\partial \underline{u}}{\partial t} + (\nabla \underline{u}) \underline{u} \right) - \eta \nabla^2 \underline{u} = -\nabla \pi$$

Non-dimensionalize with generic quantities to define standard dimensionless parameters.

- Dependent variables: $\underline{u}, \pi$
- Independent variables: x, t
- Parameters: $\rho \left[\frac{M}{L^3} \right] \quad \mu \left[\frac{M}{LT} \right] \rightarrow \nu = \frac{\mu}{\rho} \left[\frac{L^2}{T} \right]$
+ Geometry, BC, IC

Use parameters to scale the variables:

$$\underline{u}' = \frac{\underline{u}}{u_c} \quad \pi' = \frac{\pi}{\pi_c} \quad \underline{x}' = \frac{x}{x_c} \quad t' = \frac{t}{t_c}$$

Note: $\underline{u}', \pi', \underline{x}'$ and t' are dimensionless

substitute into governing equations

$$\rho \frac{v_c}{t_c} \frac{\partial \underline{u}'}{\partial t'} + \rho \frac{v_c^2}{x_c} (\nabla'_x \underline{u}') \underline{u}' - \frac{\mu v_c}{x_c^2} \nabla'^2_x \underline{u}' = - \frac{\pi_c}{x_c} \nabla'_x \pi'$$

Choose to scale to accumulation term

$$\underbrace{\frac{\partial \underline{u}'}{\partial t'}}_{\Pi_1} + \underbrace{\frac{v_c t_c}{x_c} (\nabla'_x \underline{u}') \underline{u}'}_{\Pi_2} - \underbrace{\frac{\nu t_c}{x_c^2} \nabla'^2_x \underline{u}'}_{\Pi_3} = - \underbrace{\frac{\pi_c t_c}{x_c \rho v_c} \nabla'_x \pi'}_{\Pi_3}$$

where $\nu = \frac{\eta}{\rho}$ kinematic viscosity
 $\nu \left[\frac{L^2}{T} \right]$ momentum diffusivity

Three dimensionless groups $\Rightarrow$ define time scale

$$\Pi_1 = \frac{v_c t_c}{x_c} = 1 \Rightarrow \text{advective scale} \quad t_c = t_A = \frac{x_c}{v_c}$$

$$\Pi_2 = \frac{\nu t_c}{x_c^2} = 1 \Rightarrow \text{diffusive scale} \quad t_c = t_D = \frac{x_c^2}{\nu}$$

Use Π_3 to define pressure scale

$$\Pi_3 = \frac{\pi_c t_c}{x_c \rho v_c} = 1 \Rightarrow \pi_c = \frac{x_c \rho v_c}{t_c}$$

Choose a diffusive time scale $t_c = \frac{x_c^2}{\nu}$

Dimensionless N-S equation:

$$\frac{\partial \underline{v}}{\partial t} + \frac{v_c x_c}{\nu} (\nabla'_x \underline{v}') \underline{v}' - \nabla'^2_x \underline{v} = - \nabla'_x \pi'$$

$\Rightarrow$ one remaining dim. less group

$$\text{Re} = \frac{v_c x_c}{\nu}$$

Reynolds number

ratio of time scales: $\text{Re} = \frac{t_D}{t_A} = \frac{\text{diffusive}}{\text{advective}}$

Hence we have (dropping primes)

$$\frac{\partial \underline{v}}{\partial t} + \text{Re} (\nabla \underline{v}) \underline{v} = \nabla^2 \underline{v} - \nabla \pi$$

$\Rightarrow$ only one governing dimensionless parameter: Reynolds number

For viscous flow of glacier:

$$\rho = 10^3 \frac{\text{kg}}{\text{m}^3} \quad v_c = 100 \frac{\text{m}}{\text{yr}} \sim 10^{-6} \frac{\text{m}}{\text{s}}$$

$$\mu = 10^{14} \text{ Pa s} \quad x_c = 10^2 \text{ m (thickness)}$$

$$\text{Re} = \frac{v_c x_c \rho}{\mu} = \frac{10^{-6+2+3}}{10^{14}} = 10^{-1-14} = 10^{-15} \ll 1$$

$\Rightarrow$ advective momentum transport is negligible

For $Re=0$ we have

$$\boxed{\begin{aligned}\frac{\partial \underline{v}}{\partial t} - \nabla^2 \underline{v} &= -\nabla \pi \\ \nabla \cdot \underline{v} &= 0\end{aligned}}$$

Equations are linear!

But is it worth resolving diffusive timescale?

$$t_D = \frac{x_c^2 \rho_0}{\mu} = 10^{4+3-14} \text{ s} = 10^{-7} \text{ s}$$

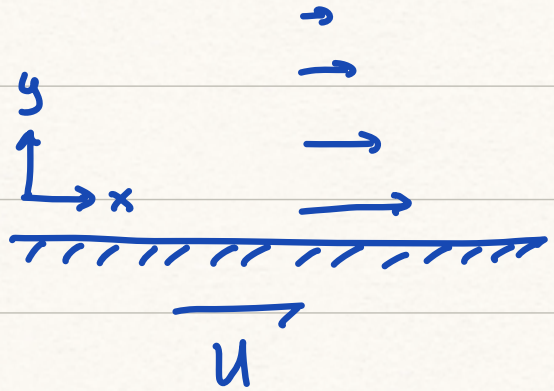
This is very short compared to 100 years of glacier response. Not worth resolving transients.

Can't eliminate transient term because we scaled to it $\Rightarrow$ scale to diffusion term.

$\Rightarrow$ next lecture

Rayleigh's problem

- Semi-infinite half-space
- Stationary fluid
- Impulsively started plate with velocity U .



$$v_c = U \rightarrow Re = \frac{U x_c \rho_0}{\mu} \ll 1 \Rightarrow U \ll \frac{\mu}{x_c \rho_0}$$

But what is x_c ? Not obvious

Redimensionalize assuming $Re \ll 1$

$$\frac{\partial \underline{v}}{\partial t} - \nu \nabla^2 \underline{v} = -\nabla \pi \quad \& \quad \nabla \cdot \underline{v} = 0 \quad \underline{v} = \begin{pmatrix} u \\ w \end{pmatrix}$$

Simplify the equations:

Domain is infinite in x but $|\pi| < \infty \Rightarrow \frac{\partial \pi}{\partial x} = 0$

Flow is horizontal: $\underline{v} = \begin{pmatrix} u \\ w \end{pmatrix} \Rightarrow w = 0$

From continuity: $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial y} = 0 \Rightarrow \frac{\partial u}{\partial x} = 0 \Rightarrow u = u(y)$

$$\nabla^2 \underline{v} = v_{i,jj} \underline{e}_i \quad i, j \in \{1, 2\}$$

$$= \begin{pmatrix} v_{1,11} & v_{1,22} \\ v_{2,11} & v_{2,22} \end{pmatrix} = \begin{pmatrix} \cancel{u_{xx}} & u_{yy} \\ \cancel{w_{xx}} & \cancel{w_{yy}} \end{pmatrix} = \begin{pmatrix} u_{yy} \\ 0 \end{pmatrix}$$

Substituting we have

$$x\text{-mom.: } \frac{\partial u}{\partial t} - \nu \frac{\partial^2 u}{\partial x^2} = 0$$

$$y\text{-mom.: } 0 = -\frac{\partial \pi}{\partial y}$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}} \quad \text{with } u(0, y) = 0 \quad u(t, 0) = U$$

This is identical to heating a semi-infinite rod from the end.

Problem has self-similar solution in

$$\eta = \frac{y}{\sqrt{4\nu t}} \quad \text{and} \quad u(y, t) = U f(\eta)$$

where $\sqrt{4\nu t}$ takes role of char. length that depends on t .

$$\text{derivatives: } \frac{\partial \eta}{\partial t} = -\frac{1}{2} \frac{\eta}{t} \quad \frac{\partial \eta}{\partial y} = \frac{1}{\sqrt{4\nu t}}$$

The derivatives of u transform as:

$$\frac{\partial u}{\partial t} = U \frac{df}{d\eta} \frac{\partial \eta}{\partial t} = -\frac{U}{2} \frac{\eta}{t} \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = U \frac{d^2 f}{d\eta^2} \left(\frac{\partial \eta}{\partial x} \right)^2 = \frac{U}{4\nu t} \frac{d^2 f}{d\eta^2}$$

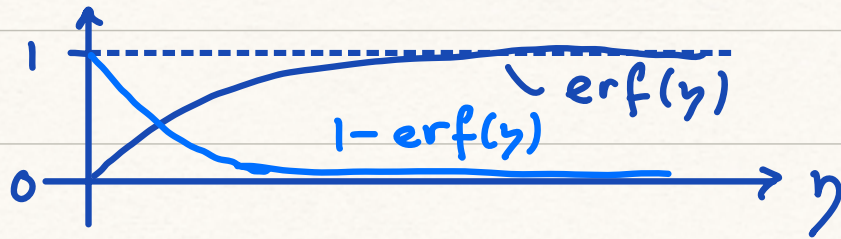
substituting into PDE:

$$\frac{d^2 f}{d\eta^2} + 2\eta \frac{df}{d\eta} = 0 \quad \text{with } f(\eta=0) = 1$$

Reduce PDE in y and t to ODE in η

Solution: $f(\eta) = 1 - \text{erf}(\eta)$ (Gauss)

where $\text{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-\xi^2} d\xi$ error function



Resubstituting for $f = \frac{u}{U}$ and $\eta = \frac{y}{\sqrt{4\nu t}}$

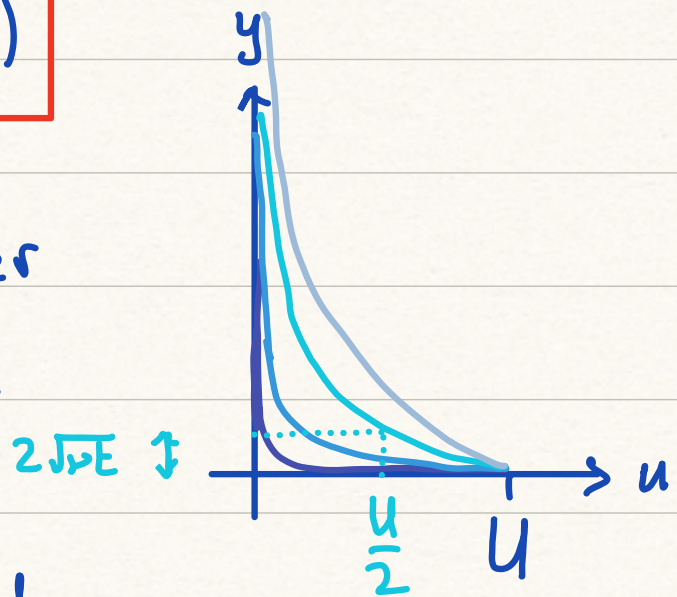
$$u(y,t) = U \left(1 - \text{erf} \left(\frac{y}{\sqrt{4\nu t}} \right) \right)$$

Diffusive boundary layer

where momentum added

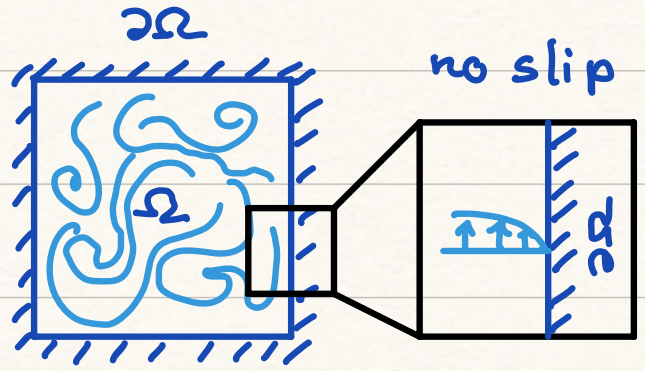
by boundary penetrates
into the quiescent fluid.

$\nu = \frac{\mu}{\rho_0}$ is Diffusion coefficient.



Kinetic Energy of Fluid Motion

Dissipation of kinetic energy in a closed domain with no-slip boundaries.



$\Rightarrow$ Energy is lost by boundary

Change in kinetic energy

$$\frac{d}{dt} K(t) = \int_{\Omega} \frac{1}{2} \rho \frac{d}{dt} |\underline{u}|^2 dV = \int_{\Omega} \rho \underline{\dot{u}} \cdot \underline{u} dV$$

from Navier-Stokes Eqs: $\rho_0 \underline{\dot{u}} = \mu \nabla_x^2 \underline{u} - \nabla \pi$

$$\frac{d}{dt} K(t) = \int_{\Omega} (\mu \nabla^2 \underline{u} - \nabla \pi) \cdot \underline{u} dV$$

$$\Rightarrow \frac{dK}{dt} = \mu \int_{\Omega} \nabla^2 \underline{u} \cdot \underline{u} dV - \mu \int_{\Omega} \nabla \pi \cdot \underline{u} dV$$

$\Rightarrow$ show second term is zero

$$\nabla \cdot (\pi \underline{u}) = \underline{u} \cdot \nabla \pi + \pi \nabla \cdot \underline{u}$$

$$\int_{\Omega} \nabla \pi \cdot \underline{v} \, dV = \int_{\Omega} \nabla \cdot (\pi \underline{v}) \, dV = \int_{\partial \Omega} (\pi \underline{v}) \cdot \hat{n} \, dA$$

$\underline{v} = 0$ on $\partial \Omega$

$$\Rightarrow \frac{dK}{dt} = \mu \int_{\Omega} \nabla^2 \underline{v} \cdot \underline{v} \, dV = ?$$

$$\begin{aligned} \nabla^2 \underline{v} \cdot \underline{v} &= v_{i,jj} \cdot v_i = (v_{i,j} v_i)_{,j} - v_{i,j} v_{i,j} \\ &= \nabla \cdot (\nabla^T \underline{v} \underline{v}) - \nabla \underline{v} : \nabla \underline{v} \end{aligned}$$

$$\frac{dK}{dt} = \mu \int_{\Omega} \nabla \cdot (\nabla^T \underline{v} \underline{v}) \, dV - \mu \int_{\Omega} \nabla \underline{v} : \nabla \underline{v} \, dV$$

$$= \mu \int_{\partial \Omega} (\nabla^T \underline{v} \underline{v}) \cdot \hat{n} \, dA - \mu \int_{\Omega} \nabla \underline{v} : \nabla \underline{v} \, dV$$

$$\frac{dK}{dt} = -\mu \int_{\Omega} \nabla \underline{v} : \nabla \underline{v} \, dV$$

Poincaré Inequality

$$\|\underline{v}\|_{\Omega} \leq \lambda \|\nabla \underline{v}\|_{\Omega} \quad \text{for } \underline{v} = 0 \quad \partial \Omega \quad \lambda > 0$$

using standard inner product

$$\int_{\Omega} |\underline{v}|^2 \, dV \leq \lambda \int_{\Omega} \nabla \underline{v} : \nabla \underline{v} \, dV$$

Notice λ has units of L^2 and scales with area of Ω .

$$\frac{dk}{dt} \leq -\mu \lambda \int_{\Omega} |\underline{v}|^2 dV = -\underbrace{\frac{\mu \lambda}{\rho_0} \int_{\Omega} \rho |\underline{v}|^2 dV}_K$$

$$\Rightarrow \boxed{\frac{dk}{dt} \leq -\frac{\mu \lambda}{\rho_0} k}$$

Solve by separation of parts

$$\frac{dk}{k} \leq -\frac{\mu \lambda}{\rho_0 \lambda} dt = -\alpha dt$$

$$\ln k \leq -\alpha t + c_0$$

$$k \leq c_1 e^{-\alpha t}$$

Initial condition $k(0) \leq c_1 = k_0$

$$\boxed{k(t) \leq k_0 e^{-\frac{\nu}{\lambda} t}}$$

$\nu = \frac{\mu}{\rho}$ mom. diffusivity

$\lambda =$ geometric factor